
Practical 7 : Introduction to Machine Learning using Scikit-learn

Program :

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error
```

```
#-----
#Step 1: Create Dataset
#-----
data = {
    "StudyHours": [4, 6, 3, 4, 5, 6, 7, 5, 9, 8, 2, 3, 4, 5, 6],
    "Marks": [35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 42, 48, 52, 58, 63]
}
df=pd.DataFrame(data)
print(df)
```

	StudyHours	Marks
0	4	35
1	6	40
2	3	45
3	4	50
4	5	55
5	6	60
6	7	65
7	5	70
8	9	75
9	8	80
10	2	42
11	3	48
12	4	52
13	5	58
14	6	63

```
#-----
#Step 2: Select StudyHours And Target
#-----

x=df[["StudyHours"]]
y=df["Marks"]

#-----
#Step 3: Split Data
#-----

X_train, X_test, y_train, y_test = train_test_split(
    x, y,
    test_size=0.2,
    random_state=42)

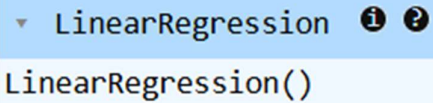
print("\nTraining Data Size :",len(X_train))
print("\nTesting Data Size :",len(X_test))
```

```
Training Data Size : 12
```

```
Testing Data Size : 3
```

```
#-----  
#Step 4: Create Model  
#-----  
model=LinearRegression()
```

```
#-----  
#Step 5: Train Model  
#-----  
model.fit(X_train,y_train)
```



LinearRegression

LinearRegression()

```
#-----  
#Step 6: Predict Results  
#-----  
predictions=model.predict(X_test)  
print("\nActual Marks:",list(y_test))  
print("\nPredicted Marks:",list(predictions))
```

```
Actual Marks: [80, 48, 35]
```

```
Predicted Marks: [np.float64(68.47345132743364), np.float64(46.90265486725
```

```
#-----  
#Step 7: Evaluate Model  
#-----  
mae=mean_absolute_error(y_test,predictions)  
mse=mean_squared_error(y_test,predictions)  
print("\nMean Absolute Error:",round(mae,2))  
print("\nMean Squared Error:",round(mse,2))
```

```
Mean Absolute Error: 9.61
```

```
Mean Squared Error: 132.35
```

```
#-----  
#Step 8: Make New Prediction  
#-----  
StudyHours=[[9]]  
predictions_marks=model.predict(StudyHours)  
print("\nPredicted Marks for 9 Study Hours:")  
print("Expected Marks",round(predictions_marks[0],2))
```

```
Predicted Marks for 9 Study Hours:  
Expected Marks 72.79  
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739:  
warnings.warn(
```